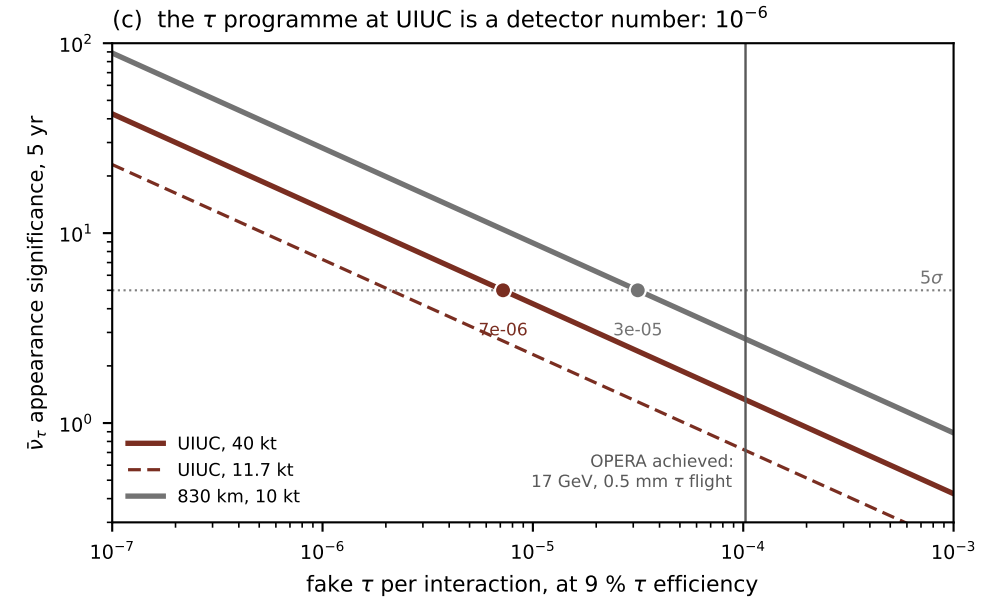
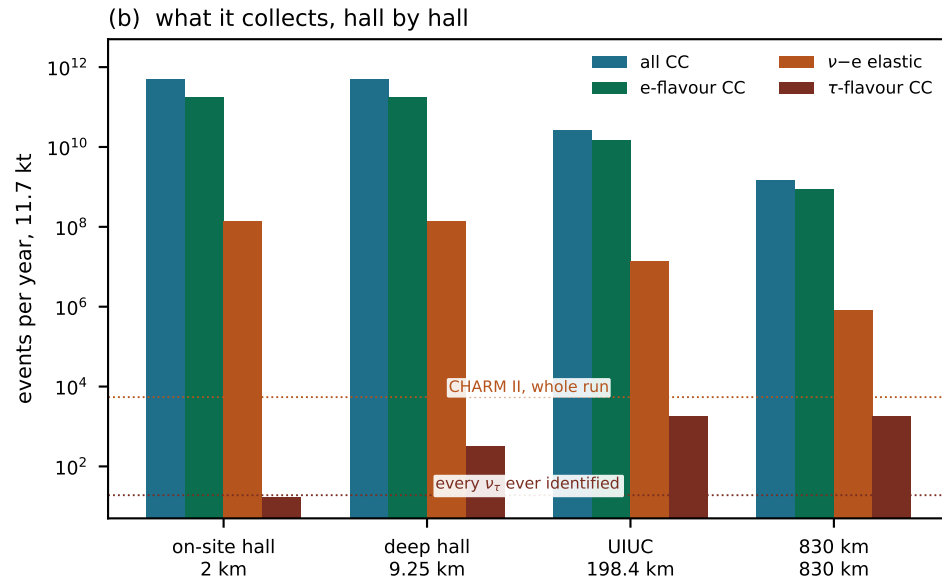
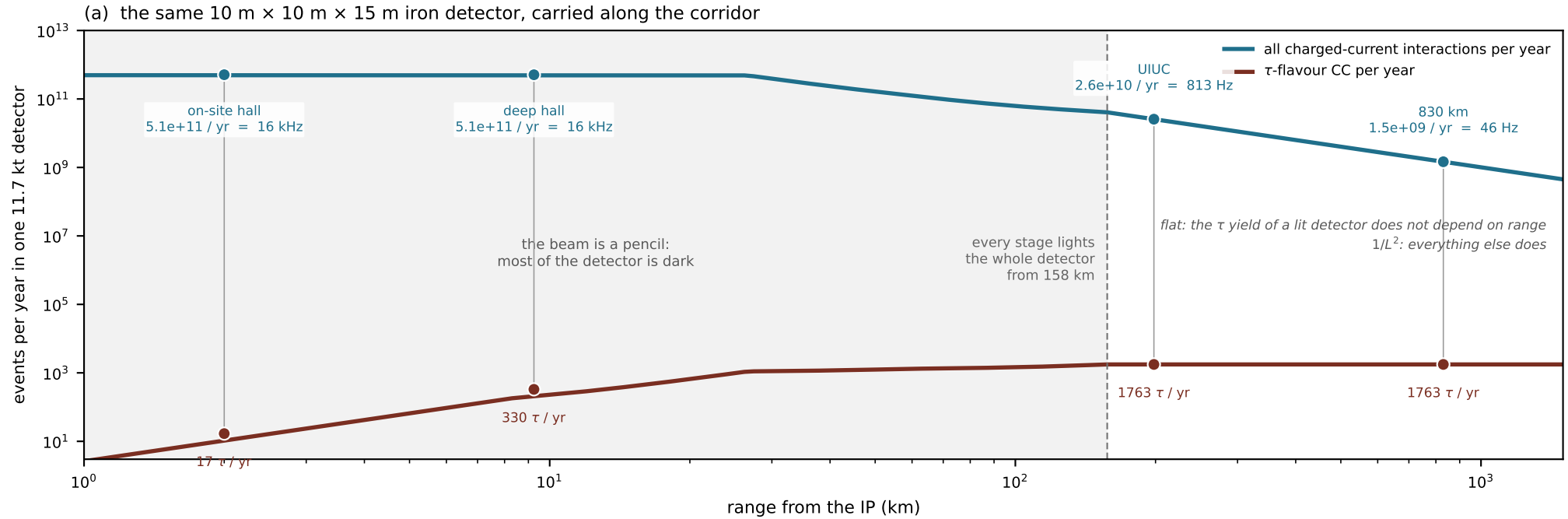


The case for a detector at UIUC

closer in the beam is a pencil and a kilotonne detector is dark; farther out the τ yield is the same and everything else is 17× rarer. 198 km is the first fully-lit hall and the last high-rate one.



Per-tonne on-axis rates from tools/nutau_sites.py, scaled by the fraction of the detector the plume lights (πr_{50}^2 for the $1/\gamma$ pencils, $2\pi\sigma^2$ for the divergence-smeared store, at each stage's mean energy). ν -e elastic: $\sigma = kE$ with the textbook k per species, on iron.

Significance uses the chirp as a weighted fit with OPERA's 9 % τ efficiency (physics_case.py). CHARM II PLB 335 (1994) 246: 5 429 ν -e events, $\sin^2\theta_W = 0.2324 \pm 0.0083$. Near halls are on the north line (μ^- decays): CC rates identical, τ rates 1.6× the south values shown.